

Realizable Nearlinear-Gap Hardness for Small Monotone Formulas

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<https://github.com/Quanyra/realizable-cmmsa-hardness>

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Abstract

For every sufficiently large fixed leaf bound L , we prove randomized polynomial-time many-one NP-hardness of realizable Collection Minimum Monotone Satisfying Assignment with weight gap $L^{1-o(1)}$ and NO satisfaction threshold $o(1)$. The corresponding bounded-advice learning statement follows through Hirahara–Nanashima’s transfer. This establishes the $\epsilon=0$ strengthening asked in Section 7 of their June 23, 2026 revision. We combine their star-projection compilation with Minzer–Zheng’s Grassmann PCP machinery, an altered repetition parameter order, explicit advice-posterior and zoom-out estimates, a maximal-extension threshold ladder, and finite-list completeness repair. Every leaf/advice bound is fixed independently of input length; the polynomial exponent may depend on it. The contribution is the nearlinear exponent in the realizable regime, with earlier realizable hardness explicitly credited below.

Keywords: hardness of approximation; monotone formulas; realizable learning; PCPs.

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1 Problems, conventions, and results

A monotone formula is an AND/OR formula with positive variable leaves; repeated occurrences count separately. $\mathcal{F}[L]$ denotes formulas with at most L leaves. Constants arising from empty branches may be simplified away. For x in $\{0, 1\}^N$ and positive rational weights w_i with $\sum_i w_i=1$, write $w(x) = \sum_{i:x_i=1} w_i$. An instance of $\text{Gap}_\sigma^{\epsilon,\gamma} \mathcal{F}[L]\text{-CMMSA}$ is an indexed nonempty explicit list $F_1, \dots, F_M$, those weights, and a rational budget $0 < s \leq 1$. The promises are:

- YES: some x with $w(x) \leq s$ satisfies at least a $1-\epsilon$ fraction of the list.
- NO: every x with $w(x) \leq \sigma s$ satisfies strictly less than a γ fraction.

Indices count with multiplicity. Weights and budgets have polynomial bit encodings and inverse-polynomial lower bounds in the number of variables. This is the normalized-weight restriction of HN Definitions 1.7 and 4.1. The distributional version DMMSA specifies the formula distribution by a sampling circuit. Our final output is an explicit power-of-two list, also represented by its exact uniform sampler. Realizable means $\epsilon=0$.

A randomized polynomial-time many-one reduction maps each input to one output instance, and for each input satisfies the appropriate YES or NO promise with probability at least $2/3$. All

randomness and output lengths are polynomially bounded. The theorem asserts one such uniform reduction on input instances for each fixed L , with no common polynomial exponent asserted across L . Limits as L grows compare this family of separately fixed reductions.

Theorem 1 (Realizable CMMSA). *There are functions σ_L in the positive integers and γ_L in $(0,1)$, with*

$$\frac{\log \sigma_L}{\log L} \longrightarrow 1, \quad \gamma_L \longrightarrow 0. \quad (1.1)$$

such that, for every sufficiently large fixed integer L , $\text{Gap}_{\sigma_L}^{0,\gamma_L} \mathcal{F}[L]$ -CMMSA is NP-hard under randomized polynomial-time many-one reductions. The output weights and budget can be required to share a polynomial-magnitude common denominator.

For completeness we specify the learning model. Fix the universal machine U used by HN. In their Definition 3.11, a distribution on $\{0,1\}^n$ has advice complexity at most a if a function α mapping random strings of length 2^{2^n} to strings of length at most a makes $U(\alpha(r); r, n)$ halt on every r , with failure-output probability at most $1/4$ and conditional output statistical distance at most 2^{-2^n} from the distribution. The advice can depend on r and need not be computable. This parameter is not the length of a conventional fixed sampler description. For labeled examples use the total encoded sample length in this definition.

$\text{Gap}_{\sigma}^{\epsilon,\gamma} \text{Learn}[a]$ takes unary length and program-budget parameters $1^n, 1^s$ and an example oracle E on $\{0,1\}^n$ times $\{0,1\}$, promised to have sampling advice complexity at most a . Its YES case has a program of description length at most s running in linear time and agreeing with the labels with probability at least $1-\epsilon$. Its NO case bounds the agreement of every program of description length at most σs by $1/2+\gamma/2$. The NO assertion does not impose an efficient running-time restriction on those programs. We use HN's machine and encoding convention; the universal overhead below is the associated constant c_U . Our reduction outputs an explicit sampling circuit, so the hardness also holds with that circuit supplied in place of only oracle access.

Corollary 1 (Realizable learning). *For every sufficiently large fixed integer advice cap a , put*

$$\begin{aligned} L &= a - 2\lceil \log_2(a+1) \rceil - c_U, \\ \sigma_a^{\text{learn}} &= \lfloor 0.49\sigma_L \rfloor, & \gamma_a^{\text{learn}} &= 5\gamma_L. \end{aligned} \quad (1.2)$$

Then $\text{Gap}_{\sigma_a^{\text{learn}}}^{0,\gamma_a^{\text{learn}}} \text{Learn}[a]$ is NP-hard under randomized polynomial-time many-one reductions. For large a the parameters are in their required domains, $\sigma_a^{\text{learn}} = a^{1-o(1)}$, and $\gamma_a^{\text{learn}} = o(1)$. In YES instances the linear-time program has agreement exactly one; in NO instances all programs within the expanded description budget have agreement at most $1/2+\gamma_a^{\text{learn}}/2$.

These are approximation-hardness results under randomized reductions. They give no complexity-class separation, one-way-function construction, linear-factor hardness, or growing-parameter reduction. No assertion about general improper PAC learning or circuit size is made.

2 Relation to previous work

HN revision 1 [3] proves the nearlinear exponent with vanishing nonzero YES error and asks its realizable strengthening separately from its superconstant-parameter question. Hirahara's FOCS 2022 work [2] already proves realizable learning hardness; HN explicitly records the corresponding realizable CMMSA factor L^α for a constant $\alpha>0$, with threshold $1/L^\alpha$. The advance here is the

nearlinear exponent, not the first realizable hardness theorem. The source contracts are the corrected June 23, 2026 revision, not the withdrawn superconstant regime of an earlier version. [3, 2].

The MZ [5] higher-query construction gives soundness nearly R^{-m} with $m+1$ queries but its displayed main theorem chooses the alphabet after both positive errors. Our proof changes the construction's parameter order; substituting an inverse-alphabet completeness error into that main theorem alone would be circular. MZ24's [6] two-query result, even granting its alphabet-before-completeness abstract formulation, does not provide this higher-arity contract: independent two-query repetition spends two queries per repetition. Likewise a perfect-completeness dictatorship test is not by itself this unconditional star-projection PCP. The targeted literature assessment found no exact subsumer, but does not certify novelty or priority. [5, 6, 1].

Slack variables, AND products, concentration, weight rounding, secret sharing, and the star compilation are established techniques. The new argument is their quantitative composition with the modified repetition, actual advice posterior, conditional rank/zoom-out control, and descending threshold ladder. We import the local Grassmann theorems and the outer hardness machinery rather than reprove them. The STOC 2026 full text of MZ was not accessible in the assessment; claims about its numerical details are confined to the accessible arXiv v1, also linked by a coauthor.

3 Notation and imported contracts

All vector spaces are over $\mathbb{F}_2$. $\text{Grass}(E, d)$ is the set of d -dimensional linear subspaces of E , with uniform distribution unless stated otherwise. $\begin{bmatrix} n \\ a \end{bmatrix}_2$ is the Gaussian binomial coefficient counting a -subspaces of an n -space. SD is total variation distance (one half the L1 distance). $\text{Zoom}[Q, W]$ is the uniform d -subspace L with $Q \subseteq L \subseteq W$, where $d=2h$. The symbol L for a subspace is local to the Grassmann argument; the scalar L in Theorem 1 always denotes the final leaf bound. Similarly, a denotes advice dimension in that argument and the final advice cap only in Corollary 1. The final asymptotics use base-two log, and $\ln$ is natural log. The parameters m, h, ρ, J are leaf-query count, inner dimension parameter, rational decoding slack, and repetition count. Set $r=10m/\rho$, choosing rational ρ with this value integral. Thus r is a fixed integer bound on advice dimensions and zoom-out codimensions; the local decoder gives $a+c \leq r$.

The proof uses the following established inputs at their stated parameter ranges. Precise public versions and bibliographic links are in the bibliography. The contracts are explicit mathematical imports; none is a locally formalized theorem.

Imported contract. Outer hardness and game (MZ Theorem 3.1, Section 3.2, Claim 3.2). There is an absolute NO value $s_0 < 1$ for 3-Lin, while its fixed positive YES error may be arbitrarily small. Use the bounded-occurrence form with no pair of variables in more than one equation. In the smooth repeated game choose J equations independently; in each triple keep all coordinates with probability $1-\beta$, otherwise a uniform single coordinate. Both provers receive the same r random vectors on the retained coordinates, extended by zero to the first question. They answer assignments; acceptance requires agreement on retained variables and satisfaction of the J equations. With fixed NO gap $\eta=1-s_0$, its value is at most $2^{-\Omega(\eta^2 2^{-r} \beta J)}$; its honest failure is at most J times the outer YES error. In particular the soundness constant does not depend on that later YES error.

Imported contract. Star construction and transport (MZ Section 3.3, Lemmas 3.3–3.4). Keep tuples U of J disjoint equations with no cross-equation pair of variables occurring together in any equation. View U as a $3J$ -space and let H_U be the span of its equation indicator

vectors, of dimension J . Labels at $L+H_U$, with $\dim L=2h$ and L transverse to H_U , are linear functions satisfying the equation right-hand sides on H_U ; there are $R=2^{2h}$ labels. Center labels are linear functions on a transverse $2(1-\rho)h$ -subspace K . Two leaf vertices are equivalent when $L+H_U+H'_U=L'+H_U+H'_U$. The imported lemmas give an equivalence relation and unique side-condition-preserving *label* transport. The test chooses U uniformly, K uniformly transverse, m independent transverse L_i containing K , then a uniform representative from each leaf's equivalence class. It transports each queried *label* back and tests restriction to K against the center *label*. This is a weighted star-projection $(m+1)$ -CSP, with alphabets at most R ; all choices are finite and enumerable in $N_{\text{outer}}^{O_m(J)}$. Clique-consistency selection and the collision estimate from Section 5.4.1 preserve the value, up to a vanishing Gaussian-binomial collision loss bounded above by our explicit estimates.

Imported contract. Local decoder (MZ Theorem 4.2). For fixed small $\rho>0$ and sufficiently large integral h with integral test dimensions, transverse tables obeying the above side conditions and test density at least $2^{-2(1-1000\rho)hm}$ have dimensions a,c with $a+c\leq 10m/\rho$ such that at least 2^{-6h^2} of the uniform a -subspaces Q admit W containing $Q+H_U$, of codimension c , and a linear g on W respecting H_U , with agreement at least $C=2^{-2(1-1000\rho^2)h}/5$ on the conditioned Grassmann space. Transversality losses are charged explicitly below.

Imported contract. Maximal-pair counting (MZ Definition 5.4; MZ24 revision 1 Theorem 5.26). For a total table of linear functions on $\text{Grass}(V, 2h)$, a pair (W, g) is $(B, 1/5)$ -maximal relative to Q if it agrees with the table on at least B of $\text{Zoom}[Q, W]$, and no proper extension W' with a linear extension of g agrees on at least $B/5$ of $\text{Zoom}[Q, W']$. Specialize MZ24 with field $\mathbb{F}_2$ and its $\delta=\rho/m$, fixing its subsidiary constants as in Lemma 5.24 before taking h large. For $\dim V\geq 2^h$, $\dim Q\leq 10m/\rho$, $\text{codim } W\leq 10m/\rho$, and $B\geq 2^{-2(1-(\rho/m)^3)h}$, the number of these maximal pairs is at most $B^{-2} 2^{O_{m,\rho}(h)}$. We use the stronger cutoff $B\geq 2^{-2(1-\rho^3)h}$, which implies this one. Thus we require no advice range proportional to J from the broader printed MZ Theorem 5.5. Counting does not assert extension at a single lowered threshold; the threshold ladder below supplies that step.

Imported contract. Covering (KMS [4], Definition 4.5, Lemmas 4.6–4.7). For the independent triple-deletion sampler with $2^d \beta\leq 1/8$, basic d -subspace distance is at most $\beta \sqrt{J} 2^{d+4}$. For advice dimension $a<d$, except for at most $\sqrt{\beta} J^{1/4}$ of uniform a -subspaces Q , the distance after conditioning on containing Q is at most $\sqrt{\beta} J^{1/4} 2^{d+5}$. The condition is on the joint sampler, changing the distribution of V . These covering lemmas are unconditional results proved in KMS Section 8; they do not require its historical combinatorial hypothesis or UGC.

Imported contract. Compilation (HN Lemma 4.6). For a star constraint $e=(y; x_1, \dots, x_m)$, let $\pi_{e,i}$ map its leaf alphabet to the center. Its acceptance is $\pi_{e,i}(\text{label}(x_i))=\text{label}(y)$ for every i . Give a vertex u occurrence weight $\lambda(u) = \frac{\mathbb{E}_e[\mathbf{1}[y=u] + \sum_i \mathbf{1}[x_i=u]]}{m+1}$, discarding zero-occurrence vertices. Set $\Lambda = \sum_u \lambda(u) |\Sigma_u|$, $w(z_{u,b})=\lambda(u)/\Lambda$, and $s=1/\Lambda$. If X_e is the set of distinct leaf vertices, let $P_{e,x,b}$ contain labels at x whose projections at every occurrence of x equal b . Output the monotone formula

$$F_e = \bigvee_{b \in \Sigma_y} \left[z_{y,b} \wedge \bigwedge_{x \in X_e} \left(\bigvee_{a \in P_{e,x,b}} z_{x,a} \right) \right]. \quad (3.1)$$

Empty ORs are false. Repeated leaf consistency is imposed in P , not discarded. The formula has at most $(m+1)R$ leaves. Value at least $1-\tau$ gives a weight- s assignment satisfying at least $1-\tau$ of these formulas. CSP value at most ζ implies every assignment of weight at most $\frac{1}{8}(5/8)^{1/(m+1)}\zeta^{-1/(m+1)}$ satisfies at most $3/4$ of them.

Imported contract. Secret sharing and learning (HN Theorem 3.5 and Lemma 5.1). Formula leaf bound L supplies a polynomial-time secret-sharing scheme with total share length at most L . For a circuit-sampled distribution of such formulas, positive inverse-polynomial rational weights and budget, and parameters ϵ in $[0,1]$, γ in $(0,1]$, $\sigma \geq 1$, the randomized transfer preserves ϵ , changes γ to 5γ and σ to 0.49σ , and gives advice at most $L+2\lceil\log_2(L+1)\rceil+c_U$. Its YES guarantee holds for every transfer random string; its NO guarantee fails with only negligible probability. A sufficiently large polynomial length parameter must make all weighted string lengths integral and satisfy the description/soundness bounds. Our common-denominator step supplies this requirement. Integer gap factors are obtained by rounding down.

We now prove the parameter extension and the remaining composition in full. Lemma 1 gives the finite-list repair; Sections 6–10 supply the parameter and reduction arguments for Theorem 1 and Corollary 1. The local input theorems above retain their usual external-proof status.

4 Explicit-list exception lemma

Lemma 1 (Finite-list completeness repair). *Input: $M \geq 1$ monotone formulas $F_1, \dots, F_M$ on N variables, each with at most L leaves; positive rational weights w_i summing to one; positive rational budget $s \leq 1$; integer $\sigma \geq 8$; rational $0 < \Gamma < 1/2$; and rational ϵ with $\epsilon \geq 0$ and $\epsilon \sigma \leq \Gamma/2$. The promises are:*

- *YES: some assignment of weight at most s satisfies at least $1-\epsilon$ of the list, counting indices with multiplicity.*
- *NO: every assignment of weight at most σs satisfies strictly less than Γ of the list.*

Set $\lambda = \sigma s / \Gamma$. Introduce M fresh variables e_j and replace F_j by $F_j \text{ OR } e_j$. Give original variables raw weights w_i and every e_j raw weight λ/M ; divide every weight by $1+\lambda$. Set

$$t = \frac{s + \lambda\epsilon}{1 + \lambda}, \quad \sigma_{\text{new}} = \lfloor \sigma/4 \rfloor, \quad \Gamma_{\text{new}} = 2\Gamma. \quad (4.1)$$

The resulting list has at most $L + 1$ leaves per formula. In the YES case an assignment of normalized weight at most t satisfies every formula. In the NO case every assignment of normalized weight at most $\sigma_{\text{new}} t$ satisfies strictly less than Γ_{new} of the list.

Proof. In the YES case, take an original witness and enable exactly the exception bits for its failed list indices. Its raw weight is at most $s + \lambda\epsilon$, and it satisfies every augmented formula. No witness or failed-index set is needed to compute the reduction: these choices establish existence.

In the NO case, any augmented assignment of normalized weight at most $\sigma_{\text{new}} t$ has raw weight at most

$$\lfloor \sigma/4 \rfloor (s + \lambda\epsilon) \leq \frac{3\sigma s}{8}. \quad (4.2)$$

Its original-variable part therefore satisfies less than Γ of the original list. Its enabled exception bits occupy at most

$$\frac{3\sigma s/8}{\lambda} = \frac{3\Gamma}{8} \quad (4.3)$$

of list indices. The augmented acceptance fraction is strictly less than $11\Gamma/8$, hence strictly less than 2Γ . This proves the claimed perfect-completeness gap reduction. Normalization cancels in both budget comparisons, and $t \leq 1$ because $s \leq 1$ and $\epsilon \leq 1$.

The output has $N+M$ variables, M formulas and at most $L+1$ leaves per formula. Construction copies the list, adds M OR gates, and performs rational arithmetic on input rationals. It is polynomial in the actual input and output bit lengths. For fixed σ, Γ , inverse-polynomial lower bounds on w_i and s imply the same for the normalized weights, t , and λ/M . No SAT, optimum, or counting oracle is used. Applying this directly to an exponential implicit distribution would be invalid; sampling precedes it. $\square$

5 Why the repetition parameters are changed

The alphabet of the MZ star construction is $R=2^{2h}$. Its displayed choice $J=2^{100h^2}$, $\beta=\log_2 \log_2 J/J$ has $\beta J=O(\log h)$, so its outer-game estimate alone does not give an exponent proportional to h^2 . We use $J=2^{2^{Ah^2}}$, $\beta=A h^2/J$. This observation concerns only arXiv 2510.23991v1; it is not a disproof of its main theorem or a claim about the uninspected STOC 2026 text. The argument below proves the needed parameter order using the local imported contracts.

6 Parameter extension: posterior and zoom-out argument

This section supplies the parameter modification rather than treating an inverse-alphabet completeness error as part of a black-box theorem. All logarithms here are base two ($\ln$ denotes the natural logarithm). Fix m and a positive rational ρ . Use the local decoding and maximal-pair counting contracts stated above (MZ Theorem 4.2 and the narrower MZ24 Theorem 5.26 specialization). Their parameters r and the constants in their bounds depend on m, ρ ; their ambient space may be enlarged. Choose h sufficiently large relative to these fixed parameters and a multiple of b_m , the denominator of the selected rational ρ . Then $2(1-\rho)h$ and all test dimensions are integers. All subspaces below are over $\mathbb{F}_2$, U consists of J disjoint triples, and $\dim(U)=3J$. The side-condition space H_U has dimension J .

Write a for an advice dimension, $0 \leq a \leq r$, and c for a zoom-out codimension, $0 \leq c \leq r$. Write $d=2h$. In the outer sampler each triple retains all three coordinates with probability $1-\beta$, and otherwise retains a uniformly chosen single coordinate. Thus the number of dropped blocks D is $\text{Bin}(J, \beta)$, and $\dim(V)=3J-2D$. Let P be the uniform distribution on a -subspaces Q of U . Let P' be the marginal obtained by first sampling V , then sampling a uniform a -subspace of V .

For $J=2^{2^{Ah^2}}$ and $\beta=A h^2/J$, the advice statistical distance obeys

$$\text{SD}(P, P') \leq \delta_{\text{adv}} := \beta \sqrt{J} 2^{a+4}. \quad (6.1)$$

This is smaller than 2^{-100h^2} for sufficiently large h . Likewise the KMS zoom-in exceptional fraction

$$\delta_{\text{zoom}} := \sqrt{\beta} J^{1/4} \quad (6.2)$$

and the conditional distance $\delta_{\text{zoom}} 2^{d+5}$ are smaller than 2^{-100h^2} , after increasing the fixed lower bound on h . The condition $2^d \beta \leq 1/8$ is satisfied. Increasing A does not cause a circular choice: A depends only on fixed decoding/outer constants, and h is chosen afterward.

Set $T=h^4$ and $\zeta=2^{-30h^2}$. The binomial tail has the explicit bound

$$\Pr[D > T] \leq \left(\frac{eAh^2}{T} \right)^T \leq 2^{-100h^2} \quad (6.3)$$

for sufficiently large h (round T to an integer if needed). Except on a P' -mass at most 2^{-70h^2} of Q , Markov's inequality gives $\Pr[D > T \mid Q] \leq \zeta$. Passing from P' to P adds at most δ_{adv} to this exceptional mass. Further, the set where $P'(Q) < P(Q)/2$ has P -mass at most $2\delta_{\text{adv}}$, by the definition of statistical distance. Intersect these good sets with the KMS good zoom-in set. The excluded P -mass is at most

$$\delta_{\text{zoom}} + 3\delta_{\text{adv}} + 2^{-70h^2} < 2^{-20h^2}. \quad (6.4)$$

Fix a Q outside this exceptional set. For any V with $D \leq T$, Bayes' rule and the Gaussian binomial formula give

$$\begin{aligned} \frac{\Pr[V \mid Q]}{\Pr[V]} &= \frac{\mathbf{1}[Q \subseteq V]}{\left[\begin{smallmatrix} \dim V \\ a \end{smallmatrix} \right]_2 P'(Q)} \\ &\leq 2 \frac{\left[\begin{smallmatrix} 3J \\ a \end{smallmatrix} \right]_2}{\left[\begin{smallmatrix} 3J-2D \\ a \end{smallmatrix} \right]_2} \leq 8 \cdot 2^{2aT}. \end{aligned} \quad (6.5)$$

The last inequality follows by factoring out 2^{2aD} in the product formula; the remaining finite products are bounded by an absolute constant because $3J-2D$ is much larger than a . This is a density bound for the actual posterior, not an assumption that V remains unconditioned.

Now let $W=W(Q)$ be any subspace containing Q , of codimension $c \leq r$. Once Q is fixed, W is fixed before V is sampled. Express W as the kernel of a rank- c matrix. Failure of $\text{codim}_V(W \cap V) = c$ means that some nonzero linear combination of its rows has support entirely outside V . For each of the at most $2^c - 1$ combinations, select one nonzero coordinate. That coordinate is removed with probability at most β . A union bound therefore gives the unconditional estimate $(2^c - 1)\beta$, uniformly in W . Applying the posterior density bound on $D \leq T$, and then the tail bound, gives, for this same Q ,

$$\begin{aligned} \Pr[\text{codim}_V(W \cap V) \neq c \mid Q] &\leq 8 \cdot 2^{2aT} (2^c - 1)\beta + \zeta \\ &\leq 2\zeta. \end{aligned} \quad (6.6)$$

The last inequality holds for large h because J is double exponential in h^2 . The estimate is valid for every $W(Q)$ of the permitted codimension; there is no union over all W , no assumption that W is chosen independently of Q , and no use of an unconditional rank estimate after conditioning.

The KMS distribution conditioned on Q is exactly the posterior needed here. Indeed, the probability a uniform d -subspace L of V contains Q is

$$\mathbf{1}[Q \subseteq V] \frac{\left[\begin{smallmatrix} d \\ a \end{smallmatrix} \right]_2}{\left[\begin{smallmatrix} \dim V \\ a \end{smallmatrix} \right]_2}. \quad (6.7)$$

The numerator is independent of V , so conditioning the joint (V, L) sampler on Q subset L induces the same V posterior as sampling Q uniformly inside V . This also explains why an unconditioned outer draw is invalid.

For the zoom-out, put $b=d-a$ and $p_0=2^{-cb}$. On the rank-stable event, the additional probability of L subset W , given V and Q , is

$$p_V = \frac{\left[\frac{\dim V - a - c}{b} \right]_2}{\left[\frac{\dim V - a}{b} \right]_2} = p_0(1 + O(2^{-J/2})). \quad (6.8)$$

This product estimate is uniform: $\dim(V) \geq J$ always, while a, c, d are $O_{m,\rho}(h)$. Under the uniform U sampler the corresponding probability is at least $p_0/2$ for large h . Conditioning two distributions at distance Δ on an event of probability at least $p_0/2$ changes their distance by at most $O(\Delta/p_0)$. Apply this with the KMS conditional distance $\Delta = \delta_{\text{zoom}} 2^{d+5}$.

Finally compare the posterior mixture with the mixture additionally weighted by p_V . Rank-unstable V have posterior mass at most 2ζ ; their contribution to the normalized weighted mixture is $O(\zeta/p_0)$. On the remaining V , p_V/p_0 differs from one by $O(2^{-J/2})$. Hence for any event concerning L , the difference between these two mixtures is

$$O(2^{-J/2} + \zeta/p_0). \quad (6.9)$$

Both this expression and Δ/p_0 are much smaller than 2^{-10h^2} , since $p_0 \geq 2^{-2rh}$. Therefore agreement with a decoded function on a uniform L satisfying $Q \subseteq L \subseteq W$ transfers to the actual posterior experiment with additive error below 2^{-10h^2} . This supplies the conditional-on- W step with the correct V distribution. It also supplies the rank stability required when the two provers extend their functions.

6.1 Decoder probability is measured in the same joint law

The outer advice consists of shared random vectors in V (MZ Section 3.2), and both provers take the span of the same first a vectors after independently guessing a . Given linear independence, the span is uniform among a -subspaces of V . The probability of dependence is at most $\sum_{i=0}^{a-1} 2^{i-\dim V} \leq 2^{r-J}$. Thus the actual vector-advice law and the ideal experiment V then uniform Q subset V differ in statistical distance by at most 2^{r-J} . This error is negligible compared with every success term below; no alteration of the outer game is needed.

Fix the leaf-label assignment T_1 after the imported clique-consistency selection, with T_2 the center-label assignment. A good first-prover question U is one whose conditional inner-test density reaches the local decoder's threshold. If the composed value exceeds $R^{-(1-\xi)m}$, averaging after clique selection gives a mass $2^{-O_m(h)}$ of such U : keep questions of conditional density at least half the surviving total density. The slack $1000\rho \leq \xi/4$ and sufficiently large h absorb this factor of two and the collision loss in the decoder threshold.

Fix a good first-prover question U . In the ideal joint experiment the marginal law of Q is P' , and the conditional law of V is exactly the posterior analyzed above. The favorable advice set has P -mass at least $2^{-6h^2} \cdot 2^{-20h^2} \cdot 2^{r+1-2J}$. Passing to P' subtracts δ_{adv} , so its P' -mass is at least 2^{-8h^2} . For each such Q the expected agreement under the actual posterior V is at least $C \cdot 2^{-10h^2}$, hence at least $C/2$. Since agreement is in $[0,1]$, the posterior probability that it is at least $C/4$ is at least $C/4$. Subtracting the rank-failure probability 2ζ leaves at least $C/8$ for large h .

These two probabilities multiply within the actual joint experiment: $P'(\text{good } Q)$ times $\Pr[\text{good } V \mid Q]$. They are never converted to an unconditional V success probability for a fixed Q . In particular, the factor $8 \cdot 2^{2ah^4}$ was used solely to upper-bound a rank-failure event; its reciprocal is not a loss in the decoded success bound. The independent-vector correction subtracts at most 2^{r-J} from the final joint success, again negligible. The threshold-ladder strategy below supplies the maximal-extension step.

6.2 Total tables and transverse subspaces

The source vertices L plus H_U require $L \cap H_U = \{0\}$. For each U , extend the induced table to all d -subspaces by assigning the zero linear function on otherwise undefined entries. The local decoding Theorem 4.2 is applied with its stated side-condition/transversality hypothesis. Its resulting agreement on the full conditioned Grassmann space, or equivalently the agreement after charging the exceptional entries, loses a negligible amount quantified here.

If Q intersect $H_U = \{0\}$ and W contains $Q + H_U$ with codimension at most r , a uniform d -subspace L between Q and W fails transversality only if L/Q meets $(H_U + Q)/Q$ nontrivially. A union bound over nonzero vectors of this J -dimensional image bounds this probability by

$$2^{J+d-\dim W+1} \leq 2^{-2J+d+r+1}. \quad (6.10)$$

This follows from $\Pr[z \text{ in a uniform } b\text{-subspace of an } n\text{-space}] = (2^b - 1)/(2^n - 1)$, and applies with an extra factor at most two if a preceding distribution was already conditioned on transversality. It is negligible compared with C .

For each second-prover question V fix an arbitrary legitimate completion $U_0(V)$ containing V . Define its total table on every d -subspace L of V by $T_1[L + H_{U_0}]|L$ when transverse, and by zero otherwise. This definition depends only on V and the fixed global assignment table. Whenever L is transverse to both H_U and H_{U_0} , the source clique-consistency relation identifies the two values. No equality is asserted on other entries.

Here are explicit charges for those other entries in the actual joint law for a fixed original U . On $D \leq T$, $\dim(V) \geq 3J - 2T$. For uniform Q subset V , the probability Q meets $H_{U_0(V)}$ nontrivially is at most $2^{a+J-\dim(V)+1} \leq 2^{-2J+2T+r+1}$. This bound is valid although U_0 is selected after V . Including the tail gives joint probability at most $2^{-100h^2} + 2^{-2J+2T+r+1}$. By Markov, except on P' -mass at most 2^{-69h^2} , its posterior probability for fixed Q is at most ζ . Passing to P adds δ_{adv} . Remove these Q from the good advice set; the total removed advice mass still fits the bound 2^{-20h^2} .

For remaining Q, V with $D \leq T$, rank stability, and Q disjoint from both side-condition spaces, take L uniform between Q and W intersect V . For either side-condition space the same quotient union bound gives nontransversality probability at most

$$2^{J+d-(\dim V-r)+1} \leq 2^{-2J+2T+d+r+1}. \quad (6.11)$$

For H_{U_0} not contained in W intersect V use its intersection with that space; its dimension is at most J , which only improves this bound. Thus replacing either total table by the actually defined transverse values costs at most twice this bound on good V , plus the posterior tail and exceptional probabilities $O(\zeta)$. These are below 2^{-10h^2} . Enlarge the finite lower bound on h to absorb all such additive errors simultaneously. In the subsequent agreement ledger one may use expected agreement $C/2$, good- V agreement $C/4$, and good- V probability $C/8$ as already budgeted. No undefined table entry or unproved identity on nontransverse L is used.

6.3 Explicit threshold ladder for maximal extensions

A single arbitrarily lowered threshold does not guarantee maximality. Instead the second prover guesses j uniformly from $\{0, \dots, r\}$ and uses $B_j = C/(4 \cdot 5^j)$. For a favorable V , begin with $(W(Q) \cap V, g)$ restricted to $W(Q) \cap V$, which has codimension at most r and agreement at least $B_0 = C/4$. At stage j , if the pair is $(B_j, 1/5)$ -maximal, stop. Otherwise Definition 5.4 supplies a proper extension of the pair with agreement at least $B_j/5 = B_{j+1}$. Replace the pair by that extension. Each replacement strictly decreases codimension, so this process stops by stage r , at latest at the whole space, where no proper extension exists.

Thus at one of these $r+1$ thresholds a maximal pair exists whose function extends the first prover's restriction. The second prover enumerates all maximal pairs of codimension at most r at the guessed threshold and chooses uniformly; this is a strategy for a value bound, not a polynomial algorithm used by the reduction. The maximal-pair counting contract bounds each such list by $B_j^{-2} \cdot 2^{O_{m,\rho}(h)} = 2^{O_{m,\rho}(h)}$. Its threshold hypothesis holds: the local agreement is $C = 2^{-2(1-1000\rho^2)h}/5$, so

$$\frac{B_r}{2^{-2(1-\rho^3)h}} = \frac{2^{2(1000\rho^2-\rho^3)h}}{20 \cdot 5^r} \geq 1 \quad (6.12)$$

for sufficiently large h (take $\rho < 1$). Codimension is at most the source local bound $r \leq 10m/\rho$; the advice dimension is also within the narrower MZ24 counting contract. This costs the additional constant factor $1/(r+1)$, with no single-threshold existence assumption. The first prover takes its decoded linear function g on $W(Q)$, which respects the equation side conditions, and extends it uniformly to U . The second prover extends its selected maximal-pair function uniformly to V . For the successful choice, that function extends g restricted to $W(Q) \cap V$. Rank stability gives $\text{codim}_V(W(Q) \cap V) = c$, where $c = \text{codim}_U W(Q) \leq r$. The restriction to V of a uniform extension of g to U is uniform among the 2^c extensions of this common restriction: the quotient map $V/(W(Q) \cap V) \rightarrow U/W(Q)$ is an isomorphism. Thus it agrees with the second prover's chosen extension on V with probability $2^{-c} \geq 2^{-r}$. The first prover's assignment satisfies all outer equations because its extension preserves g on H_U . This proves the claimed random-extension success cost.

7 Completing the modified PCP parameter order

Fix $m \geq 2$ and a desired exponent slack $\xi > 0$; set $\rho = \xi/4000$, shrinking it further if necessary to meet the imported local theorems. Use the same star-query construction as MZ, with the new J, β above. The following table checks every ambient/repetition-dependent loss used in its decoding argument. Local Grassmann decoding and the maximal-zoom-out counting theorem are imported; the numerical parameter extension is argued here.

Use	Control under the replacement
Ambient dimensions in local decoding	$3J$ and $\dim(V) \geq J$ exceed every fixed lower bound depending on h ; H_U has dimension J and the same disjoint-triple structure.
Duplicate clique choices	The source Gaussian-binomial comparison decreases exponentially in J ; for large h the union over m^2 pairs is at most 2^{-J} .
Lucky advice meeting H_U	Its uniform probability is at most 2^{r+1-2J} .
Advice/zoom-in covering	The bounds above and their exceptional Q sets are less than 2^{-20h^2} , below the local decoding lucky mass 2^{-6h^2} .
Zoom-out and rank steps	The posterior calculation above gives error below 2^{-10h^2} , below the decoded agreement $C \geq 2^{-O_{m,\rho}(h)}$.
Number of maximal zoom-outs	The imported bound is $2^{O_{m,\rho}(h)}$ for the relevant agreement and codimension; its constants do not depend on J .
Random extensions	Their agreement cost is at most a factor 2^r , since codimensions are bounded by r .

Consequently the source's local-to-outer decoding strategy, with these errors retained instead of dropped, has success at least $2^{-C_*h^2}$ for some fixed $C_* = C_*(m, \xi)$, whenever the composed test

has value above $R^{-(1-\xi)m}$. To see the scale explicitly, the factors are a test-density term $2^{-O_m(h)}$, a constant dimension-guess factor, lucky-advice mass at least 2^{-8h^2} , decoded agreement $2^{-O_{m,\rho}(h)}$, a list-choice factor $2^{-O_{m,\rho}(h)}$, and extension probability at least 2^{-r} . The exponent slack between ρ and ξ absorbs the fixed multiplicative losses and the 2^{-J} clique loss. No factor depending exponentially on J is paid in this decoded success probability.

The fixed outer NO gap gives, by the outer-game bound, success at most $2^{-\kappa\beta J}$, with $\kappa > 0$ depending on the fixed advice size and outer NO gap, not on the outer YES error. Choose an integer

$$A > \frac{C_* + 10}{\kappa}. \quad (7.1)$$

Then this is at most $2^{-(C_*+10)h^2}$, contradicting the decoded success. Conditioning on legitimate equation tuples changes it by at most a factor two, as ensured by padding below. Thus the modified PCP has soundness at most $R^{-(1-\xi)m}$, where $R=2^{2h}$.

Only after fixing m, ξ, h, J, β do we fix any desired positive constant PCP completeness error τ . Select the outer 3-Lin YES error

$$\epsilon_1 \leq \frac{\tau}{100(m+1)J}. \quad (7.2)$$

The starting hardness theorem allows every fixed positive ϵ_1 with an absolute NO gap. In particular, this does not require exactly satisfiable linear equations to be NP-hard. Its reduction exponent may depend on ϵ_1 , which is allowed because every parameter here is a constant independent of the input length.

To account for discarded illegitimate tuples, take sufficiently many disjoint copies of the outer instance so their probability a is at most $\min(\tau/100, 1/4)$. The bound is $O(J^2/N_{\text{outer}})$ with a constant from the bounded occurrence condition. Copies preserve the value and degree bound. The number required depends only on the already fixed parameters; for an input of length n the padded size is $O(n + N_{\min})$, still polynomial. Conditioning on legitimate U multiplies a failure probability by at most $1/(1-a)$. The sampler's marginal for each clique-resampled U'_i is uniform over legitimate U : the initial vertex is uniform, and uniform resampling within its equivalence class preserves that measure. All relevant spaces have the same dimensions and number of transverse extensions.

An honest global labeling satisfies the composed constraint whenever the equations in the original U and all m resampled U'_i are satisfied. Union bounding gives failure at most

$$\frac{(m+1)J\epsilon_1}{1-a} \leq \frac{\tau}{75} < \tau. \quad (7.3)$$

Using all $m+1$ blocks is deliberately conservative; no equality between a single-block bound and the multi-query acceptance event is needed. This establishes the required parameter order: for each fixed m, ξ and each sufficiently large h , every fixed positive τ is available without increasing R . J and the source input-size threshold can be very large, and all are charged as fixed constants.

The construction and enumeration cost is $N_{\text{outer}}^{O_m(J)}$ with constants depending on these choices, plus the starting outer reduction. It is not a polynomial with a uniform exponent when L, h or τ vary with input length. No superconstant-parameter hardness claim is made.

8 From the modified PCP to an explicit realizable list

Use HN's star-projection compilation and its occurrence weights, retaining repeated leaf-variable consistency as in its Lemma 4.6. It gives formulas with at most $(m+1)R$ leaves, a normalized honest

budget s , and a constant NO satisfaction bound $3/4$ up to weight σs , where we may safely take the explicit integer

$$\sigma = \left\lfloor \frac{R^{1-1/m}}{16} \right\rfloor, \quad \xi = \frac{1}{m^2}. \quad (8.1)$$

For implementation this floor is computed by integer root comparison; no irrational number is an encoded weight. The constant $1/16$ is smaller than the coefficient in the source's list-decoding bound. For sufficiently large R , $\sigma \geq 8$. Set $q = \lfloor \sqrt{m} \rfloor$ and take an AND of q independently sampled base formulas. Put

$$\Gamma = 2(3/4)^q, \quad \epsilon = \frac{\Gamma}{16\sigma}, \quad \tau = \frac{\epsilon}{4q}. \quad (8.2)$$

The modified PCP above permits this τ after R has been fixed. The AND-product YES failure is at most $q \tau = \epsilon/4$; the NO satisfaction fraction is at most $(3/4)^q = \Gamma/2$. The local leaf bound is $q(m+1)R$.

Let N_0 be the number of monotone variables. Materialize the finite sampling support if necessary; for fixed parameters it has polynomial size in the outer input. Its rational probabilities and occurrence weights have polynomial bit descriptions. A positive atom is a product of a fixed number of inverse-polynomial choices, so positive variable weights and s are inverse-polynomially bounded. Adding rational numbers may enlarge a common denominator, which is handled separately below.

Construct a bounded-random-bit sampler whose distribution has statistical distance at most $\epsilon/8$ from the exact product distribution. One implementation enumerates its S atoms and rounds cumulative rational probabilities to a dyadic grid with at least $\log_2(8S/\epsilon)$ bits. Charge the support enumeration and rational arithmetic. This is polynomial for fixed L ; it is not an oracle for an exponentially large support.

Choose a natural P with $1/\epsilon \leq P$; ϵ is positive. Since ϵ is rational, $P = \lceil 1/\epsilon \rceil$ can be computed by integer division and remainder on its numerator and denominator. Use the conservative count

$$T = 32(N_0 + 11)P^2, \quad M = 2^{\text{clog}_2 T}, \quad \frac{N_0 \ln 2 + \ln 12}{2(\epsilon/8)^2} \leq T \leq M < 2T. \quad (8.3)$$

Here $\text{clog}_2 T$ is the least natural exponent whose power of two is at least T . The constructor uses only natural arithmetic. The displayed analytic bound follows from $\ln 2 \leq 1$, $\ln 12 \leq 11$ and $1/\epsilon \leq P$; it is a proved comparison, not a real-number computation required by the sampler. This one count already gives sampling success at least $5/6$, and therefore also the required $2/3$. It is the least qualifying power of two for T , not necessarily for the exact logarithmic threshold. For fixed L the previously chosen positive rational ϵ is fixed, so this choice of P is constant with respect to the outer input, and $M < 64(N_0 + 11)P^2$ bounds the list length. Support enumeration and rational arithmetic costs still have to be charged as above. The formal count and probability bounds do not certify an encoded machine runtime.

Draw M independent formulas from this sampler. Hoeffding's inequality and a union bound over 2^{N_0} assignments imply, with probability at least $2/3$, that all empirical satisfaction fractions differ from their exact product probabilities by at most $\epsilon/4$, including the sampling-distribution error. Hence the list's YES failure is at most $\epsilon/2$, and its NO acceptance is at most $\Gamma/2 + \epsilon/4 < \Gamma$. The exception lemma may therefore use the conservative error budget ϵ . Its condition $\epsilon \sigma \leq \Gamma/2$ holds with a factor-eight margin.

Apply that lemma to this actual list. The output satisfies all constraints in its YES case, with ratio $\lfloor \sigma/4 \rfloor$, threshold 2Γ , and at most $q(m+1)R+1$ leaves per formula. Failure of the randomized reduction occurs only in the sampling event already bounded by $1/3$. The completed list has an exact uniform sampler using $\log_2(M)$ random bits. In particular, the later learning reduction sees $\epsilon=0$ exactly; it does not inherit the approximate sampler used to construct the list.

9 Rational-weight control and learning transfer

Polynomial bit lengths alone do not make all weights integral multiples of $1/\text{poly}(n)$. We supply a rounding step before any use of fixed-length strings w_i in the learning reduction. Let v_i be the final normalized weights on N' variables and t its YES budget. Choose the least power of two $D \geq 8(N'+1)/t$, set $a_i = \lceil Dv_i \rceil$, $A_0 = \sum_i a_i$, and

$$B_0 = \min\{A_0, \lceil Dt \rceil + N'\}, \quad v'_i = a_i/A_0, \quad t' = B_0/A_0. \quad (9.1)$$

A YES assignment remains feasible, since its integer weight is at most $Dt + N'$. Conversely, any assignment with v' -weight at most $(\sigma_{\text{new}}/2)t'$ has original v -weight at most

$$\frac{\sigma_{\text{new}}}{2} \frac{B_0}{D} \leq \frac{\sigma_{\text{new}}}{2} \left(t + \frac{N'+1}{D} \right) \leq \frac{9}{16} \sigma_{\text{new}} t < \sigma_{\text{new}} t. \quad (9.2)$$

Thus the NO acceptance threshold is unchanged if the gap factor is rounded down to $\lfloor \sigma_{\text{new}}/2 \rfloor$. Formula leaves and satisfaction are unchanged. D and A_0 are polynomial numeric quantities because t has an inverse-polynomial lower bound; all new weights and the budget share the polynomial common denominator A_0 . The clipping in B_0 guarantees $t' \leq 1$ and does not weaken either displayed inequality.

Represent the power-of-two list by its exact sampler. Every monotone formula with L leaves admits the source's L -bit-total secret-sharing construction. HN Lemma 5.1 explicitly allows $\epsilon=0$: it keeps that completeness error, multiplies the NO threshold by five, loses a factor 0.49 in the size gap, and uses advice bound

$$L + 2\lceil \log_2(L+1) \rceil + c_U. \quad (9.3)$$

Its length parameter can be chosen as a polynomial multiple of A_0 , so all strings of length v'_i times that parameter have integral lengths. The parameter is also increased by a polynomial factor to meet its stated description and soundness bounds. These adjustments are polynomial in the actual instance size. Perfect completeness holds for every randomness string of this transfer; its additional randomized failure probability is on the NO side. The conservative list count above already gives success at least $5/6$; combine this with the transfer's sufficiently small failure probability to obtain a randomized many-one reduction with success at least $2/3$. Finitely many small inputs can be handled by a fixed truth table, a constant cost depending on the fixed parameters.

10 Final asymptotic parameter choice

For each fixed sufficiently large target leaf bound L , consider integers $m \geq 256$ with $m \leq \sqrt{\log_2 L}$. Let $q = \lfloor \sqrt{m} \rfloor$ and

$$h(L, m) = b_m \left\lfloor \frac{\log_2((L-1)/(q(m+1)))}{2b_m} \right\rfloor, \quad R = 2^{2h}. \quad (10.1)$$

Additionally require $b_m \leq \sqrt{\log_2 L}$. Require $h(L, m)$ to exceed every fixed lower bound used above for that m and $\xi=1/m^2$, including $\sigma \geq 8$ and all numerical error inequalities. Choose the largest admissible m . Every fixed m is admissible for all sufficiently large L , so $m(L)$ tends to infinity. The finite selection does not assert a polynomial-time algorithm uniform in L ; for each fixed L it fixes the constants of a uniform polynomial-time reduction on inputs. The same fixed-parameter convention is used in the revised source.

The final leaf bound is at most L by construction. Moreover,

$$\begin{aligned} \log R &= \log L - O(\log(m+1) + b_m) = \log L - o(\log L), \\ \frac{\log \sigma}{\log L} &\longrightarrow 1, \\ \Gamma &= 2(3/4)^{\lfloor \sqrt{m} \rfloor} \longrightarrow 0. \end{aligned} \tag{10.2}$$

All floor operations, the exception lemma, weight rounding and learning transfer lose only constant factors in σ . They therefore preserve $\sigma = L^{1-o(1)}$. For the learning target with advice cap a , choose $L = a - 2\lceil \log_2(a+1) \rceil - c_U$; for large a this is positive and the advice overhead fits within a . Also L/a tends to one, preserving the exponent.

Theorem 1 follows with the explicit parameters

$$\sigma_L = \left\lfloor \frac{\lfloor \sigma/4 \rfloor}{2} \right\rfloor, \quad \gamma_L = 2\Gamma. \tag{10.3}$$

where σ , Γ , m , h and R are selected above. For sufficiently large L these are in the promised ranges. The conservative natural list count above gives sampling success at least $5/6$. Corollary 1 follows from the stated HN transfer with its residual error made at most $1/6$; the combined success is at least $2/3$. The exact sampler and complete repaired YES witness ensure that the learning error parameter is exactly zero. This completes the proofs of Theorem 1 and Corollary 1. $\square$

11 Research and review disclosure

This manuscript was developed through AI-assisted mathematical derivation, source challenge, independent agent mathematical review, dependency audit, and source/significance assessment under Quantyra Research. These are informal AI reviews, not independent human peer review or Lean verification. The source challenger contributed to the argument and is not counted as an independent mathematical verifier. Review roles, findings, and limits are disclosed in the companion review disclosures. Bibliographic versions and imported inputs are recorded in the bibliography.

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